Graphic Era HILL UNIVERSITY

End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: B. Tech. C.E and 3rd sem.
Name of the Paper: *Building Materials and Construction Equipment*
Paper Code: TCE-304

Time: 3 hours Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks) CO1

a. Differentiate between normal strength concrete and reinforced cement concrete (RCC) in terms of composition, properties, and applications.

b. Compare lightweight concrete and high-density concrete, focusing on their methods of production, properties, and specific applications.

c. Explain the differences between high-strength concrete and high-performance concrete, including their unique properties and common applications.

Q2. (2X10=20 Marks) CO2

a. Discuss the need for high-performance pavement materials and analyze how they contribute to extending the lifespan of pavements.

b. Explain the concept of energetically modified cement (EMC) and evaluate how it improves the properties of concrete mixtures.

c. Describe the advantages of using warm-mix asphalt concrete compared to traditional hot-mix asphalt, particularly in terms of energy efficiency and emissions.

Q3. (2X10=20 Marks) CO3

a. Identify the benefits of using natural fibers and composites in construction, and assess how they contribute to sustainability.

b. Discuss the advantages of using green materials such as earthen materials, bamboo, and engineered wood in sustainable construction.

c. Explain the properties and applications of fiber cement in modern construction. How does it contribute to durability and sustainability?

Q4. (2X10=20 Marks) CO4

a. Compare the main functions of dozers and scrapers in earth-moving operations, and analyze how they differ in their applications.

b. Classify the different types of loaders used in construction and examine how they vary based on application and capacity.

c. Describe the role of graders and analyze how they contribute to surface finishing and leveling in construction.

Q5. (2X10=20 Marks) CO5

a. Explain the working principles and components of asphalt paving equipment used in road construction.

b. Describe the various equipment used for placing, compacting, and curing concrete in large-scale construction projects.

c. Identify the primary types of equipment used for pumping water on construction sites and evaluate how they are selected based on site requirements.